

EDITORIAL



Immunotherapy with $\gamma\delta$ T-cells: the future is there

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Cancer immunotherapy has witnessed dramatic progress in recent years. Among the most exciting achievements are the introduction of tumor-targeting monoclonal antibodies and bispecific engagers, the development of checkpoint inhibitors unleashing powerful antitumor T-cell responses and the generation of chimeric antigen receptors (CAR) redirecting effector cells to tumor-expressed antigens [1,2]. Such therapies are preferentially effective in hematological malignancies (antibodies, bispecific engagers, CAR T-cells) or in subgroups of patients with solid tumors (checkpoint inhibitors). An additional immunotherapeutic strategy is adoptive cell therapy (ACT) with *in vitro* expanded effector cells. Various effector cells can be considered for ACT including tumor-specific cytotoxic T-cells derived from blood or from tumor-infiltrating lymphocytes [3], NK cells [4], mucosa-associated T-cells [5] and others. Patient-derived autologous ACT products like human leukocyte antigen (HLA)-restricted tumor-specific CD8 cytotoxic T-cells derived from tumor-infiltrating lymphocytes are difficult to generate and must be manufactured individually for each patient. From a translational viewpoint, it would be desirable to use effector cells which are not HLA-restricted and thus could be used across HLA barriers. This would enable production of 'off-the-shelf' ACT products from healthy donors which could be stored and made available whenever needed.

There are two immune cell populations with potent antitumor activity that do not depend on HLA-restricted antigen recognition by T-cell receptors (TCR): NK cells (which completely lack TCR expression) and $\gamma\delta$ T-cells (which express a TCR different from the HLA-restricted $\alpha\beta$ TCR of conventional CD4 or CD8 T-cells). The activation of NK cells is governed by a variety of activating and inhibitory receptors, some of which interact with HLA

class I or HLA class I-related molecules [4]. NK cells recognize and kill a variety of cancer cells and do not induce graft-versus-host disease or cytokine release syndrome. Various strategies are in clinical trials to explore the full potential of unmodified and modified NK cells including allogeneic NK cells in cancer immunotherapy [4,6].

$\gamma\delta$ T-cells constitute a numerically small subpopulation of all T-cells in the blood (but a substantially higher proportion in mucosal tissues). Based on variable TCR gene usage, there are two major subsets of $\gamma\delta$ T-cells in humans. V γ 9V δ 2 T-cells (termed V δ 2 in the following) are the dominant subset in the peripheral blood. V δ 2 T-cells can kill a variety of hematological and solid malignancies. These cells recognize small pyrophosphate molecules which are overproduced in transformed cells compared with healthy cells. Research in recent years has discovered a fascinating new principle of how such 'phosphoantigens' are actually recognized by the $\gamma\delta$ TCR: phosphoantigens bind to the intracellular domains of members of the butyrophilin (BTN) transmembrane protein family, specifically BTN3A1. In concert with other BTN members like BTN2A1, this leads to a conformational rearrangement of the extracellular BTN3A1/2A1 complex which then interacts with the $\gamma\delta$ TCR and induces cellular activation. The activation of V δ 2 T-cells does not involve interaction with HLA molecules and thus is not HLA-restricted. Therefore, it should be possible to transfer V δ 2 T-cells expanded *in vitro* from healthy donors into cancer patients. In fact, we have performed a large clinical phase I study where a total of 432 V δ 2 T-cell products were transfused into 137 patients with solid cancers. The study concluded that the transfer of expanded V δ 2 T-cells across HLA barriers is safe and does not induce graft-versus-host disease [7]. In this study and several previ-

ous, V δ 2 T-cells were selectively expanded *in vitro* with zoledronic acid, which induces accumulation of endogenous $\gamma\delta$ T-cell activating phosphoantigens in monocytes [8].

The second most frequent $\gamma\delta$ T-cell subset in humans carries the variable δ chain element V δ 1 which can pair with various V γ genes. V δ 1 T-cells are more frequently encountered in mucosal tissue like the small intestine compared with blood and comprise a substantial proportion of tumor-infiltrating $\gamma\delta$ T-cells in solid tumors. Like V δ 2 T-cells, V δ 1 T-cells do not depend on interaction with HLA molecules for TCR-mediated target cell recognition. While the nature of TCR ligands for V δ 1 T-cells is less well defined, it appears that V δ 1 T-cells recognize and kill tumor targets mainly through activating NK receptors like NKp30 and DNAM-1 [9,10]. Protocols have been established which allow for the large-scale expansion of V δ 1 T-cells (termed DOT-cells) for clinical application [9]. Taken together, the two main subsets of HLA-nonrestricted human $\gamma\delta$ T-cells (i.e., V δ 1, V δ 2) are amenable for *in vitro* expansion under clinical-grade GMP conditions and thus can be exploited for allogeneic ACT. Moreover, there are also new approaches for *in vivo* activation of tumor-reactive V δ 2 T-cells based on agonistic anti-BTN3A antibodies which selectively activate V δ 2 T-cells [11].

While both $\gamma\delta$ T-cell subsets demonstrate a potent capacity to kill a variety of tumor target cells *in vitro*, the clinical efficacy of nonmodified adoptively transferred $\gamma\delta$ T-cells has been moderate so far. However, many strategies are currently being considered to improve $\gamma\delta$ T-cell immunotherapies as summarized in recent reviews [12–15]. In the following, we highlight some of the key targets for improving $\gamma\delta$ T-cell immunotherapeutic efficacy.

1. Manufacturing process

It is important to identify healthy blood donors whose $\gamma\delta$ T-cells can reproducibly expand better than in others. This is critical when aiming to establish large cell banks of frozen $\gamma\delta$ T-cells as an ‘off-the-shelf’ product. It also remains to be determined whether donor–recipient HLA discrepancy plays a role in treatment success. Optimization of cell culture conditions including the selection of cytokines is also critical. In this context, the selection of an optimal cytokine mix is important but other factors may help to improve the cellular expansion and/or effector function. The addition of vitamin C to $\gamma\delta$ T-cell cultures substantially increases the functionality of $\gamma\delta$ T-cells [7,16], likely because vitamin C has antioxidant activity but is also a well characterized epigenetic modifier [17].

2. Targeting $\gamma\delta$ T-cells to tumor cells

Bispecific T-cell engagers (BiTE) crosslinking the CD3 part of the CD3/TCR complex with tumor-expressed antigens have been successfully introduced into clinical application, with the CD3xCD19 BiTE blinatumomab having revolutionized the treatment of acute lymphoblastic leukemia. BiTE which specifically crosslink the $\gamma\delta$ TCR (either V γ 9 or V δ 2) with tumor-expressed antigens such as the EGFR strongly enhance the killing of tumor cells expressing the relevant target molecule by V δ 2 T-cells [13,18]. Alternatively, $\gamma\delta$ T-cells (both V δ 1 and V δ 2) can be equipped with CAR constructs directed against tumor-expressed surface molecules [19,20]. This can be accomplished with allogeneic $\gamma\delta$ T-cells, thus underscoring the powerful prospect of off-the-shelf $\gamma\delta$ T-cell immunotherapy. A decisive advantage of using $\gamma\delta$ rather than $\alpha\beta$ T-cells as recipients of CAR constructs is the additional presence of the tumor-reactive endogenous TCR (in case of V δ 2 T-cells) or the additional activity of potent NK receptors like NKp30 and DNAM-1 (in case of V δ 1 T-cells).

3. Overriding the immunosuppressive tumor microenvironment

A major obstacle to successful cellular immunotherapy is the immunosuppressive tumor microenvironment. A variety of suppressive cells (like myeloid-derived suppressor cells, Treg and others) and immunosuppressive molecules (like transforming growth factor- β , galectin-3 and others) may counteract an efficient antitumor response of tumor-infiltrating T-cells but also of adoptively transferred T-cells including $\gamma\delta$ T-cells [21]. Therefore, all ACT approaches for the immunotherapy of solid tumors should also address how to increase the migration of transferred T-cells into the tumor site. In this context, the local activation of innate immune receptor pathways might create an inflammatory milieu supporting the infiltration and activation of intratumoral T-cells including $\gamma\delta$ T-cells [22]. Also, a high dose intravenous infusion of vitamin C might help to boost the *in vivo* functionality of $\gamma\delta$ T-cells [17].

Overall, $\gamma\delta$ T-cells provide a unique source of effector cells for cancer immunotherapy. They are suitable as an ‘off-the-shelf’ product and amenable to manipulation via CAR transduction or BiTE application. Obviously, the identification of the best immunotherapeutic strategies based on $\gamma\delta$ T-cells will require sophisticated clinical studies. In view of the ever-increasing number of companies focusing on $\gamma\delta$ T-cells, we are convinced that $\gamma\delta$ T-cells will soon occupy a major space in the world of cellular cancer immunotherapies.

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